

Syllabus for MATH 3220-002: Foundations of Analysis II

Instructor: Alp Uzman

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1 Staff and Meeting Hours

Instructor: Alp Uzman (uzman@math.utah.edu)

Lectures: Mon Tue Wed Fri 9:40 AM - 10:30 AM @ JFB B-1

Canvas: <https://utah.instructure.com/courses/1200080>

Office Hours: By appointment

3210-3220 Coordinator: Dragan Miličić (milicic@math.utah.edu)

2 Course Description & Prerequisites

MATH 3220 is a rigorous course in multivariable calculus, serving as a transition course from calculus to analysis. As outlined in the [general catalog](#), it is worth 4 credits and has the following prerequisites:

'C' or better in MATH 3210 AND (MATH 2250 OR MATH 2270 OR MATH 2271)

Much like its prequel MATH 3210, MATH 3220 is a proof based class, and the emphasis will be on developing "[mathematical maturity](#)", in particular on rigorous argumentation, rather than calculation of the values of certain limits, integrals or derivatives. We'll start with a review of linear algebra, and an introduction to metric spaces. Then

we'll discuss differential and integral calculus of multivariable vector valued functions. Finally we'll bring together differential and integral calculi through the formalism of differential forms.

3 Expectations

As a student, you are expected to:

- attend all lectures (four times per week),
- complete 12 problem sets,
- take 2 in-class midterm exams, and
- take a final exam.

According to the [university regulations](#), you should expect a workload of approximately 8 to 12 hours per week outside of lecture hours. You will have some familiarity with many ideas and notions we will discuss throughout the semester. However the focus on rigor can be hard to get accustomed to. Missing lectures or assignments can make it challenging to catch up. Please stay engaged and reach out if you encounter any disruptions. We're here to help, but your hard work and dedication will be essential to getting back on track!

4 Learning Objectives

Among the objectives in MATH 3220 are that you learn how to:

- ambiguate and disambiguate vectors, matrices and linear transformations,

- apply vector and operator norms in proofs and computations,
- leverage the triangle inequality in metric spaces to derive key results,
- use level and sub-level sets of metrics to describe subsets of metric spaces,
- use the foundational topological and uniform properties of metric spaces, including compactness and completeness,
- prove and apply the Banach Contraction Principle,
- prove and use foundational theorems governing continuous functions, including the Maximum Principle and Intermediate Value Theorem in the context of metric spaces,
- define and use derivatives and partial derivatives of multivariable vector-valued functions,
- prove and use the foundational theorems governing differentiable functions, including the Chain Rule, Mean Value Theorem, Taylor approximation, Inverse Function Theorem, Implicit Function Theorem and Rank Theorem,
- use critical points and extrema to describe and understand abstract landscapes,
- define Riemann integrability of multivariable vector-valued functions,
- define and use Jordan measurability and content,
- prove and apply foundational theorems governing Riemann integrable functions, including Fubini's Theorem and the Change of Variables Formula,

- define and use vector fields and differential forms,
- prove and use multivariable analogs of the Fundamental Theorem of Calculus from single variable calculus via the formalism of differential forms.

5 Tentative Schedule

The course's subject matter is divided by weeks within the **academic calendar** as follows. For your convenience the approximate corresponding sections in the main textbook are also listed. Time permitting, additional topics¹ may be covered.

Week 1 (Week of 08/18): Linear Spaces 1 (§7.1, 8.4-5) Linear spaces (aka vector spaces). Linear and affine subspaces. Linear and affine functions. Linear and affine isomorphisms. Kernel and image. Linear independence and span. Bases. Matrices. Elementary matrices. Row reduction as matrix factorization. Relativity. Dimension. Classification Theorem for Finite Dimensional Vector Spaces. Rank and Nullity. Fundamental Theorem of Linear Algebra (aka Rank-Nullity Theorem).

¹Among candidate additional topics are: cokernel and coimage, Hölder inequality, Minkowski inequality, Lipschitz and Hölder continuity, Lipschitz Inverse Function Theorem, closed curves, simple curves, length of a curve in a metric space, parameterization by arc length, series of functions, absolute and conditional convergence, Weierstrass M-Test, Littlewood-Paley decomposition for Hölder functions, Riemann Rearrangement Theorem, real analytic functions, neural networks, backpropagation and automatic differentiation, universal approximation theorems, vector fields as first-order ODEs, Existence and Uniqueness Theorem for ODEs, gradient flow (aka gradient descent), jets, converses to Taylor Approximation, Whitney Extension Theorem, Sard Lemma, Morse Lemma, Moore-Penrose pseudoinverse, singular value decomposition, Schmidt-Eckart-Young-Mirsky Theorem, deterministic Oseledets Theorem, differentiation under the integral sign (aka Leibniz Integral Rule), Poincaré Lemma, Leibniz Integral Rule for differential forms, de Rham Theorem, Brouwer Fixed Point Theorem, Helmholtz Decomposition, Riemannian and pseudo-Riemannian metrics, Hodge star operator, harmonic forms, Hodge theorem, de Rham currents, Maxwell's equations. Some of these topics may show up as part of a problem set.

Week 2 (Week of 08/25): Linear Spaces 2 (§7.1) Normed spaces. Operator norm, conorm and condition number. Multilinear functions. Symmetric and alternating multilinear functions. Determinant. Duality. Inner product spaces. ℓ^p norms. Cauchy-Bunyakowsky-Schwarz Inequality.

Week 3 (Week of 09/01): Metric Spaces 1 (§7.2-3, 8.1-2) Triangular inequality. Metrics. Metric spaces. Metric subspaces. Sequences. Convergent and divergent sequences. Continuous functions. Homeomorphisms. Uniform (aka C^0) metric and norm. Product metric spaces.

Week 4 (Week of 09/08): Metric Spaces 2 (§7.4-5, 8.2-3); Review & Midterm 1 Level and sub-level sets. Open and closed balls. Spheres. Subsequences.

Week 5 (Week of 09/15): Metric Spaces 3 (§7.2-5, 8.2-3) Compact subsets. Open and closed subsets. Neighborhoods. Interior, closure and boundary operators for subsets. Bolzano-Weierstrass Theorem. Maximum Principle. Uniformly continuous functions. Cantor Intersection Theorem. Heine-Borel Theorem. Connected and path-connected subsets. Connected and path-connected components. Intermediate Value Theorem. Cauchy sequences. Complete metric spaces. Sequences of functions. Pointwise and uniform convergence. Banach Contraction Principle. Banach and Hilbert spaces.

Week 6 (Week of 09/22): Differential Calculus 1 (§9.1-4) Derivative of a multivariable vector valued function. Directional and partial derivatives. Differentiable functions. Diffeomorphisms. Chain Rule.

Week 7 (Week of 09/29): Differential Calculus 2 (§9.1-4) C^k functions. C^k metric and norm. Linear Differential Operators. Tangent and cotangent spaces. Vector fields and differential 1-forms. Gradient and differential of a multivariable real valued function.

Week 8 (Week of 10/13): Differential Calculus 3 (§9.1, 9.4-5) Mean Value Theorem. Clairaut's Theorem. Parameterized submanifolds of Euclidean space. Orientation. Taylor Approximation.

Week 9 (Week of 10/20): Differential Calculus 4 (§9.6-7, 9.5) Inverse Function Theorem. Implicit Function Theorem. Rank Theorem. Critical points. Local and global extrema. First Derivative Test. Lagrange multipliers. Hessian. Quadratic forms. Positive and negative definiteness. Second Derivative Test.

Week 10 (Week of 10/27): Integral Calculus 1 (§10.1-3); Review & Midterm 2 Aligned rectangles. Partitions. Refinements. Upper and lower Darboux sums. Riemann sums. Unsigned definite Riemann integral. Riemann integrable functions. Characteristic (aka indicator) functions. Jordan measurable subsets. Jordan content.

Week 11 (Week of 11/03): Integral Calculus 2 (§10.4-5) Partial Riemann integration. Fubini's Theorem for Riemann integrals. Determinant as volume distortion. Change of Variables Formula for unsigned definite Riemann integrals.

Week 12 (Week of 11/10): Differential Forms 1 (§11.1-3) Velocity and unit tangent vector fields of a differentiable curve. Piecewise smooth curves. Differential 1-forms. Line integrals. Fundamental Theorem of Line Integrals (aka Gradient Theorem). Pullback of differential

1-forms. Change of Variables Formula for line integrals. k -Cells. Differential k -forms. Wedge product. Exterior differentiation. Closed and exact differential forms. Musical isomorphisms. Curl, divergence and Laplacian operators. Pushforward of k -cells. Pullback of differential k -forms.

Week 13 (Week of 11/17): Differential Forms 2 (§11.4-6) Surface integrals. Green's Theorem. Stokes' Theorem (aka Curl Theorem). Volume integrals. Gauss' Theorem (aka Divergence Theorem).

Week 14 (Week of 11/24): Differential Forms 3 (§11.7) Chains. Boundary operator for chains. Cycles and boundaries. Signed definite Riemann integrals.

Week 15 (Week of 12/01): Differential Forms 4 (§11.7); Review Fundamental Theorem of Calculus (aka Generalized Stokes' Theorem).

6 Grades

The final letter grades will be determined according to the following weights and cutoffs. The cutoffs below are not definitive; based on the overall performance it may be adjusted at the end of the semester. All such adjustments will affect all students in the section equally.

Assignment	Weight
Attendance	5%
PSets	30%
Midterm 1	20%
Midterm 2	20%
Final	25%
Total	100%

Grade	Percent
A	90%
A-	85%
B+	80%
B	75%
B-	70%
C+	65%
C	60%
C-	55%
D+	50%
D	45%
D-	40%
E	else

7 Books & Other Resources

The **main textbook** for MATH 3220 is the following:

Taylor, Joseph L.. *Foundations of Analysis*. AMS. 2012.
ISBN-13: 9780821889848.

Textual Supplements As supplementary texts the following are recommended:

- Lecture notes by Terence Tao available at

[https://www.math.ucla.edu/
%7Etao/resource/general/131bh.1.03s/](https://www.math.ucla.edu/%7Etao/resource/general/131bh.1.03s/)

- Lecture notes by Ali Ülger available at

[https://web.archive.org/web/20120703144624/
http://home.ku.edu.tr/~aulger/RealAnalysis\(AU\).pdf](https://web.archive.org/web/20120703144624/http://home.ku.edu.tr/~aulger/RealAnalysis(AU).pdf)

- Paul Dawkins' multivariable notes available at

<https://tutorial.math.lamar.edu/Classes/CalcIII/CalcIII.aspx>

Further resources will be uploaded to the Canvas webpage if needed.

Videos from Previous Classes The following annotated recordings of lectures of the two main prerequisites for MATH 3220 are recommended:

- MATH 2270 (Linear Algebra), Spring 2024:

[https://www.youtube.com/playlist?list
=PL40ydqvvvXfOBtKE5ZbyvjEOffOoreeUF](https://www.youtube.com/playlist?list=PL40ydqvvvXfOBtKE5ZbyvjEOffOoreeUF)

- MATH 3210 (Foundations of Analysis I), Spring 2025:

[https://www.youtube.com/playlist?list
=PL40ydqvvvXfPGbwcE1wT2_dHvyGykxsLo](https://www.youtube.com/playlist?list=PL40ydqvvvXfPGbwcE1wT2_dHvyGykxsLo)

8 Lectures

Attendance at all lectures is expected. Attendance at random lectures will be taken and will have an effect on the final grades. The number of lectures when attendance will be taken will also be random. A certain number of recorded attendance days may be dropped at the end of the semester.

On weeks when a midterm is scheduled (i.e. Weeks 4 and 10), first two classes will focus on topics as usual; whereas the class before the midterm is reserved for review. Likewise, last class on the last week (i.e. Week 15) will be reserved for review.

Lecture Recordings Video recordings of lectures, excluding those reserved for reviews and midterms, will be uploaded weekly to the following YouTube playlist:

https://www.youtube.com/playlist?list=PL40ydqvvyXfOkGMhOrz6AID_HcpwxnzaW.

**This is an experimental feature of the course.
Recordings may become unavailable at any time
and without notice; do not rely solely on them.**

9 Problem Sets

Twelve problem sets will be assigned throughout the semester, typically on a weekly basis, excluding weeks with midterms. Each problem set is due by 11:59 PM on the Friday following its release. Late submissions, up to 24 hours after the deadline, will be accepted with a 10% penalty. Submissions more than 24 hours late will not be accepted unless you contact the course staff with a valid excuse before

the 24-hour extension expires. A certain number of problem sets with lowest scores may be dropped at the end of the semester.

Grading Criteria The problem sets will be graded partly on completion and partly on accuracy, validity, and presentation/documentation.

Submission Details For each problem set, you are required to submit a form through Google Forms and digital copies of your work through Gradescope. Detailed submission instructions will be provided in each problem set specification.

Conflicting Instructions and Deadlines The submission deadlines and methods detailed in the problem set specifications (i.e., the PDF files embedded in the associated assignment webpages on Canvas), Canvas, and Gradescope may differ. Always adhere to the deadline and instructions in the specification document, as these will be the accurate guidelines.

10 Midterm Exams

There will be two in-class midterm exams scheduled as follows:

- **Midterm 1:** September 12 (Friday of Week 4) – covering topics from Week 1 through Week 4.
- **Midterm 2:** October 31 (Friday of Week 10) – covering topics from Week 5 through Week 10.

During the midterm exams, the use of external resources (including colleagues, books, and electronics) will not be permitted. Your exams will be digitized and uploaded to Gradescope by the staff. You will have access to view how your exams were graded.

11 Final Exam

The final exam is scheduled in accordance with the university-wide [final exam schedule](#). Details are as follows:

- **Date and Time:** December 12, from 8:00 AM to 10:00 AM.
- **Content:** The final exam will be cumulative, covering all topics from the course.

During the final exam, the use of external resources (including colleagues, books, and electronics) will not be permitted. Your exam will be digitized and uploaded to Gradescope by the staff. You will have access to view how your exam was graded.

12 The Math Center

In addition to the office hours held by the course staff, the [Math Center](#) is an excellent resource for assistance with your studies. Located in the basement connecting [JWB](#) and [LCB](#), the Math Center offers free drop-in as well as online support.

13 Accessibility

The University of Utah seeks to provide equal access to its programs, services, and activities for people with disabilities. If you will need accommodations in this class, reasonable prior notice needs to be given to the [Center for Disability & Access](#), [SSB 350](#), [\(801\) 581-5020](#). CDA will work with you and the staff to make arrangements for accommodations. All written information in this course can be made available in an alternative format with prior notification to the Center for Disability & Access.

14 Inclusivity and Safety

Title IX makes it clear that violence and harassment based on sex and gender (which includes sexual orientation and gender identity/expression) is a civil rights offense subject to the same kinds of accountability and the same kinds of support applied to offenses against other protected categories such as race, national origin, color, religion, age, status as a person with a disability, veteran's status or genetic information.

If you or someone you know has been harassed or assaulted, you are encouraged to report it to:

- **Title IX Coordinator** in the **Office of Equal Opportunity and Affirmative Action**, LAW OEO Suite, (801) 581-8365, or
- **Office of the Dean of Students**, UNION 270, (801) 581-7066.

For support and confidential consultation, contact the **Center for Campus Wellness**, SSB 350, (801) 581-7776. To report to the police, contact the **Campus Security**, (801) 585-2677(COPS). For further safety resources, see **SafeU**.

15 Academic Honesty

You are expected to adhere to University of Utah policies regarding academic honesty. This means that ultimately all work you submit must be your own, created without unauthorized assistance, and all external resources, including generative tools and computer algebra systems, must be properly cited and documented within your submissions. Any student who engages in academic dishonesty or who violates the professional and ethical standards may be subject to academic sanctions as per the University of Utah's **Student Code**.

Generative AI and Computer Algebra Systems In this course, you may use generative tools like **ChatGPT** as well as computer algebra systems such as **MATLAB** under specific guidelines. These tools are permitted to guide your understanding of problem sets and topics. When using them, you must submit logs of your interactions as part of your problem set submissions. These tools can also be used to verify calculations, but remember, computations are generally expected to be done manually unless explicitly stated otherwise. You are responsible for ensuring the validity, accuracy, and relevance of your submissions, whether or not you use these tools. While encouraged, the use of such technology is not required for success in this class. The policy governing the use of such technology is specific to MATH 3220-002; other courses and sections may have different policies.

Regret Clause If you commit an unreasonable act but bring it to the staff's attention within 48 hours of the relevant submission, sanctions may be limited to that submission only, rather than leading to further disciplinary action. This clause will not be applied in the case of repeated violations.

16 Acknowledgements

The staff used DeepSeek in retouching this syllabus for clarity and readability.